FACULTY OF COMPUTING

SEMESTER 1/20252026

SECJ 3553 - ARTIFICIAL INTELLIGENCE

SECTION 02

PROJECT:

SMART AGRICULTURE SYSTEM
(ASSIGNMENT 2)

THEME:

AGRICULTURE

LECTURER:

DR. ROLIANA BINTI IBRAHIM

GROUP MEMBERS:

NAME	MATRIC ID
IMAN ABADI BIN MOHD NIZWAN	A23CS0084
MUHAMMAD AFIQ DANIAL BIN ROZAIDIE	A23CS0117
MOHAMED ALIF FATHI BIN ABDUL LATIF	A23CS0112
DHESHIEGHAN A/L SARAVANA MOORTHY	A23CS0072

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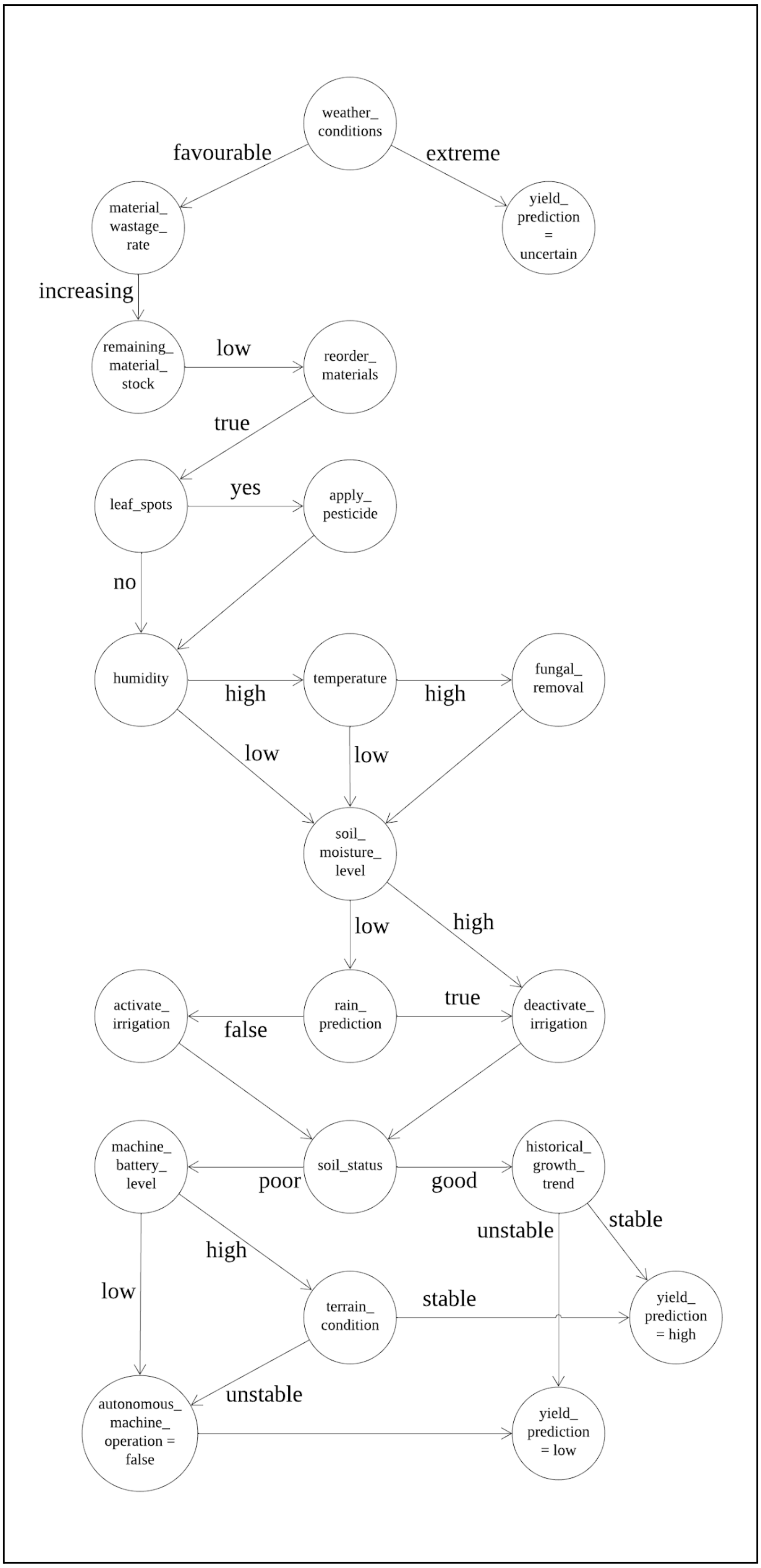
Problem Formulation By State Space Graph



Initial State = weather_conditions
Actions = monitoring_material_wastage_rate, check_remaining_material_stock, reorder_materials, check_leaf_spots, apply_pesticide, check_humidity, check_temperature, apply_fungal_removal, check_soil_moisture_level, activate_irrigation, deactivate_irrigation, predict_rain_forecast, check_soil_status, monitoring_machine, battery_level, analyse_historical_growth_trend, check_terrain_condition, predict_yeild, monitoring_autonomous_machine_operation.
Goal = yield_prediction = low **OR** yield_prediction = high **OR** yield_prediction = uncertain
Path Cost = favourable, extreme, increasing, high, low, true, false, yes, no, good, poor, stable, unstable

Description

The system begins by observing the weather conditions. If the weather becomes extreme, the system will classify the crop yield as uncertain. When the conditions are favorable, it continues monitoring wastage rates and material stock. If wastage increases and stock levels become low, the system automatically reorders materials to prevent production delays.

The system also takes care of crop health by detecting leaf spots or high humidity and temperature. When pests or fungal risks are identified, pesticide or fungal treatments are applied to protect the crops. Alongside this, the system monitors soil moisture and rainfall prediction. If the soil is too dry and no rain is expected, irrigation will be activated. Otherwise, it remains off to avoid water waste.

Soil and machinery conditions are also checked to ensure safe and efficient operations. Poor soil quality, low battery levels, or unstable terrain may result in reduced productivity or halting machine operations. Finally, by analyzing soil status and historical growth trends, the system predicts whether the crop yield will be high, low, or uncertain. Through continuous monitoring and smart decision-making, the system ensures optimal resource use and supports better farming outcomes.

Peer review on Team Working (TW)

Category	4	3	2	1
Active participation	Routinely provides useful ideas when participating in the group.	Usually provides useful ideas when participating in the group.	Sometimes provides useful ideas when participating in the group.	Rarely provides useful ideas when participating in the group.

MEMBERS/ VOTE	IMAN ABADI BIN MOHD NIZWAN	MUHAMMAD AFIQ DANIAL BIN ROZAIDIE	MOHAMED ALIF FATHI BIN ABDUL LATIF	DHESHIEGHAN A/L SARAVANA MOORTHY
IMAN ABADI BIN MOHD NIZWAN	-	4	4	4
MUHAMMAD AFIQ DANIAL BIN ROZAIDIE	4	-	4	4
MOHAMED ALIF FATHI BIN ABDUL LATIF	4	4	-	4
DHESHIEGHAN A/L SARAVANA MOORTHY	4	4	4	-